

A组

A1 ① $0 \perp 0 \geq 0 \leq \dots \leq n0$

$n+1$ 个空位, m 个男生

$$C_{n+1}^m \cdot P_m^m = \frac{(n+1)!}{m!(n+1-m)!} \cdot m! = \frac{(n+1)!}{(n+1-m)!}$$

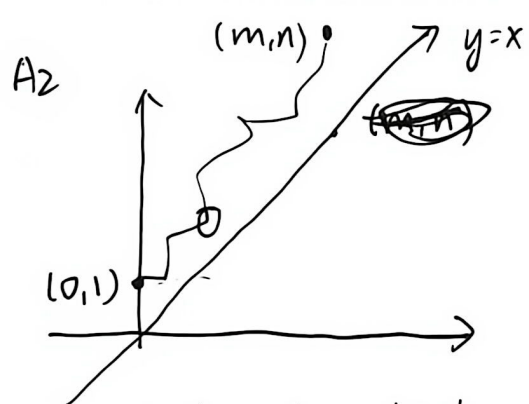
② $P_{m+1} \cdot P_n^n = (m+1)! n!$

③ 令: $P_{m+n}^{m+n} = (m+n)!$

相邻: $\frac{AB \quad BA}{m+n-1} \uparrow$

~~$2 \left[\frac{(m+n-1)!}{2} - 2(m+n-1)! \right]$~~

$(m+n)! - 2(m+n-1)! = (m+n-1)! (m+n-2)$



x轴走 m 步. y轴走 $n-1$ 步

$$\underline{y > x. \quad y \geq x+1}$$

设 $x \quad x_0. \quad y \quad y_0.$

$$(x_0, y_0+1)$$

$$y_0+1 \geq x_0+1$$

$$\underline{y_0 \geq x_0.}$$

$x: 0 \quad y: 1. \quad m \uparrow x_0 \quad n-1 \uparrow y_0$

$\underbrace{1 \quad \dots \quad 1}_{m+n-1 \text{ 个格子.}}$

$\rightarrow \quad \uparrow$

$$\begin{cases} y_0 > x_0 \\ y_0 = x_0 \\ \underline{y_0 < x_0} \end{cases}$$

$y_0 \quad y_0 \quad \dots \quad y_0$
 $n-1 \uparrow y_0$



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A3.

~~$x_i = 10 + x_j$~~

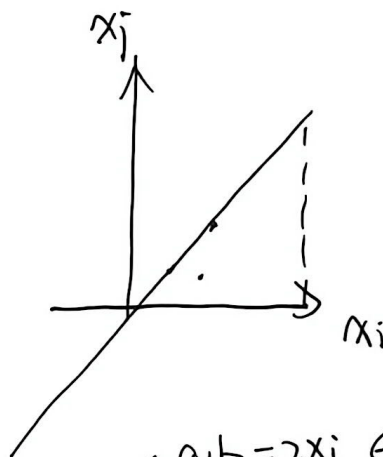
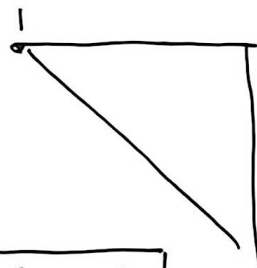
$$y_i = 10 + x_i \quad i = 1, \dots, 7$$

$$x_i \in \{0, 1, \dots, 9\}.$$

~~$\forall x_i$~~
 $\forall y_i, y_j$

$$\begin{aligned} (y_i + y_j) \bmod 10 &= (x_i + x_j) \bmod 10 \\ (y_i - y_j) \bmod 10 &= (x_i - x_j) \bmod 10. \end{aligned}$$

a_i, a_j 且 $x_i \geq x_j$



$$\begin{aligned} a &= x_i + x_j \in \{0, 18\} \\ b &= x_i - x_j \in \{0, 9\}. \end{aligned}$$

$$x_i = x_j + a \quad a \in \{0, 9\}$$

$$x_i + x_j = 2x_j + a \in \{0, 18\}.$$

反证, 且 $\begin{cases} x_i - x_j \neq 0 \\ x_i + x_j \neq 0/10 \end{cases} \bmod 10$

$x_i > x_j$ 且 $x_i < x_j < \dots < x_7$

$0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9$

$\frac{10!}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9} = 120$

$$\begin{aligned} 2x_i \bmod 10. & \quad x_i \bmod 5. \\ 2x_j \bmod 10. & \end{aligned}$$

$$2x_i = k - 10.$$

$$x_i = k \cdot 5.$$

$$0, 1, 2, 3, 5, 9$$

$$\begin{cases} a+b = 2x_i \in \{0, 18\} \\ a-b = 2x_j \in \{0, 18\} \\ b \geq 0. \end{cases}$$

$$① \ x_i + x_j = \begin{cases} 0 \\ 10 \end{cases} \quad \begin{matrix} x_i = 9 & y_i = 1 \\ 8 & 2 \\ 6 & 4 \end{matrix}$$

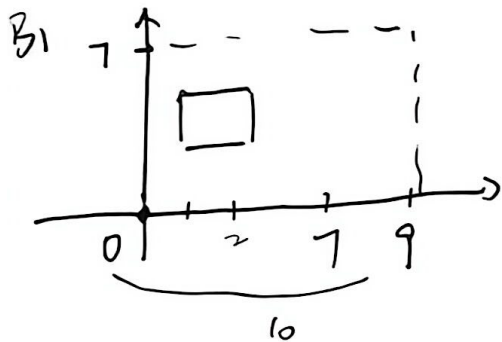
$$② \ x_i - x_j = \begin{cases} 0 \\ 10 \end{cases} \quad x$$



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B组.



① $C_{10}^2 \times C_8^2$

② 设边长为 a . ~~则~~ $a=1, \dots, 7$

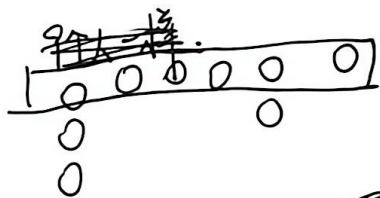
$a=1$ x : 9种 $\times$ y 轴 7种

$a=2$ x : 8种 y : 6

$\vdots$

$a=7$ 3种 y : 1种.

B2. 1 2 3 4 5 6



6个门都用: $C_9^6 \times P_6^6 \times [$

1 2 3 4 5 6 7 8 9.

~~第~~ m 个门. S_m 个方案. $m < 9$.

$m+1$ 个门. S_{m+1} ? $S_{m+1} = S_m +$

n 个人. m 个门 $n > m$.
 $S(n, m)$

$$S(n, m+1) = S(n, m) + \sum_{j=1}^n S(n-j, m) \cdot j! = S(7, 6) \cdot 6^1$$

$$S(n+1, m) = \dots$$

$$= \boxed{S(6, 6)} \cdot 6^2$$

$$= S(1, 6) \cdot 6^8 = 6^9.$$

人都一样.

$P_9^9 \times [$

$$\begin{cases} S(n, m+1) = \cancel{S(n, m)} + \sum_{j=1}^n S(n-j, m) \\ S(n+1, m) = S(n, m) \cdot m. \end{cases}$$

~~$S(9, 6)$~~ $n=9$ $m=6$.

$$S(9, 6) = S(8, 6) \cdot 6^1$$

$$= \dots$$

$$= \boxed{S(6, 6)} \cdot 6^2$$

$$= S(1, 6) \cdot 6^8 = 6^9.$$

$n \uparrow 1, m \uparrow 1$

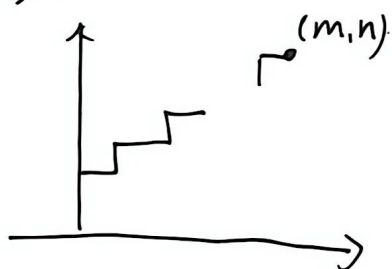
$$S(n, m+1) = S(n, m) + \sum_{j=1}^n S(n-j, m). \quad \text{降 } m.$$

$$= \sum_{j=0}^n S(n-j, m) = S(n, m) + S(n-1, m) + \dots + S(0, m)$$

$$\cancel{S(n+1, m)} = S(n+1, m+1) = \sum_{j=0}^{n+1} S(n+1-j, m) = S(n+1, m) + \dots + S(0, m)$$

$$S(n+1, m+1) - S(n, m+1) = S(n+1, m).$$

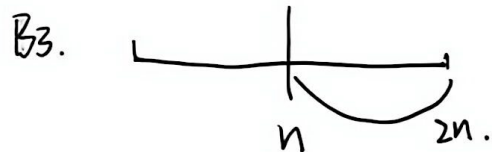
$$S(n+1, m+1) = S(n, m+1) + S(n+1, m).$$



$$C_{m+n}^m = C_{15}^6$$

$$P_9^9 \times C_{15}^6.$$





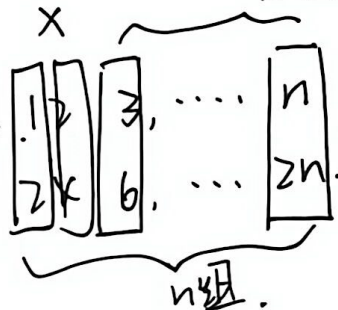
反证: 排除 1.

2. n 奇 n 偶.

$n-1$ 2

排除 2. $n-2$ 组.

只有 $2n-2$ 个可选



选 $n+1$ 个数.